

22. Edited version of K. Hentzelt: on the Theory of Polynomial Ideals and Resultants

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In what follows the question is the *general problem of elimination theory*, namely to determine the common zeros of all polynomials of an ideal of polynomials; or, what is identical with this, the zeros of any finite number of polynomials that form a basis of the ideal. At the same time a conceptual interpretation of the multiplicities that occur will emerge. The coefficients of the polynomials may be assigned to any field; the zeros then belong to the algebraically closed field derived from it. Moreover, the ideal may be assumed, without restriction of generality, to be transformed (§ 3). Then the essential result is the following:

To every ideal $\mathfrak{m}$ of polynomials one can assign a resultant form:

$$R_{\mathfrak{m}} = R^{(1)}(x_1, \dots, x_n) \cdots R^{(i)}(x_i, \dots, x_n) \cdots R^{(n)}(x_n) \equiv 0(\mathfrak{m}),$$

in such a way that $R_{\mathfrak{m}}$ vanishes for all zeros of $\mathfrak{m}$ and only for these. If $\mathfrak{n}$ is a divisor of $\mathfrak{m}$, and if the resultant forms $R_{\mathfrak{n}}$ and $R_{\mathfrak{m}}$ agree, then the ideals $\mathfrak{n}$ and $\mathfrak{m}$ also agree.¹

The second part of the main theorem shows that the resultant form supplies the zeros in characteristic multiplicity. This second part is analogous to the fact that two ideals of an algebraic number field, one of which is a divisor of the other, agree if and only if their *norms* agree; the inner reason for the two theorems is also the same.

Namely, $\mathfrak{m}$ may be conceived as a module of linear forms $\mathfrak{M}_{i-1}$, by regarding each polynomial as a linear form in the power products of $x_1, \dots, x_{i-1}$ and adjoining $x_{i+1}, \dots, x_n$ to the coefficient domain. The resultant factor $R^{(i)}$ then becomes equal to the *norm* of the corresponding fundamental module (§ 1) $\mathfrak{G}_{i-1}$ with respect to $\mathfrak{M}_{i-1}$; this also immediately gives an *interpretation of the multiplicity – product of degree and exponent of a factor of $R^{(i)}$ – by the number of linearly independent residue classes²*, a fact that until now was known only in the special case in which the resultant can be defined for indeterminate coefficients³.

For the derivation of these results, §§ 1 and 2 give theorems on modules of linear forms whose coefficients are rational integers or polynomials in one indeterminate. The connection with the ideals of polynomials introduced in § 3 is supplied by the isomorphism theorem of § 4. Section 5 gives the above-mentioned second part of the main theorem; § 7 gives the theorem on the zeros, preceded in § 6 by a general elimination theory. There, at the same time, when new indeterminates are introduced, the decomposition of the resolvent into linear forms in these indeterminates is proved; and thus, for the elimination given here, an unobjectionable proof is supplied for an assertion of Kronecker⁴.

§ 1. Modules of Linear Forms.

By integral quantities $a, b, c, \dots$ in the first two sections we shall mean rational integers or polynomials in one indeterminate with coefficients from an arbitrary, abstractly defined field P ; by linear forms $a(\xi), b(\xi), \dots$ we shall mean such forms in finitely many indeterminates $\xi_1, \dots, \xi_k$ with integral quantities as coefficients.

A *module $\mathfrak{A}$ of linear forms* is defined by the conditions: together with $a(\xi)$ and $b(\xi)$, $\mathfrak{A}$ contains the difference $a(\xi) - b(\xi)$; together with $a(\xi)$ it contains $c \cdot a(\xi)$, where c denotes an arbitrary integral quantity.

By the *rank* of $\mathfrak{A}$ we mean, as usual, the maximal number of linearly independent linear forms from $\mathfrak{A}$; by the λ -th *determinantal divisor* d_λ of $\mathfrak{A}$, the greatest common divisor of all determinants of order λ from $\mathfrak{A}$

¹Kurt Hentzelt has been missing before Dixmuiden since October 1914 and must be counted among the dead. The present article is a completely free reworking of the most essential part of his dissertation “Zur Theorie der Formenmoduln und Resultanten,” with which he received his doctorate under E. Fischer in Erlangen in the summer of 1914. This dissertation, composed entirely on the basis of his own ideas, is built up without gaps; but lemma follows lemma, all concepts are paraphrased by formulas with four and five indices, and the text is almost wholly absent, so that the greatest difficulties are presented to understanding. He did not himself get to the planned reworking. I give the work again in a purely conceptual form, by which a great simplification is obtained of proofs whose fundamental ideas throughout go back to Hentzelt, and by which, as I hope, the beauty of the work becomes evident. – The parts of the dissertation that – for a given basis – concern the question of forming the functions that occur by finitely many steps are to remain reserved for a separate publication. (E. N.)

²These latter results show their true significance when the decomposition of ideals into primary ideals is brought in; the multiplicity corresponding to a primary ideal is then defined as the number of linearly independent residue classes of the complement modulo this ideal. This is to be taken up elsewhere. (E. N.)

³Cf. Macaulay, *The algebraic theory of modular systems*, Cambridge Tracts 19 (Cambridge University Press, 1916), No. 67; also Lasker, *Zur Theorie der Moduln und Ideale*, Math. Ann. 60 (1905), pp. 20–116; p. 98. That in this case the resultant and the resultant form coincide is shown in the still unpublished theorems of Hentzelt mentioned under 1). (E. N.)

⁴The attempted proof in König, *Algebraische Größen* (Leipzig 1903, Teubner), V, § 4 – for Kronecker’s elimination theory – is known not to have succeeded. That even for the multiplicities introduced by König into Kronecker’s elimination theory the second part of Hentzelt’s main theorem does not hold is shown by Macaulay, loc. cit., p. 23; there it is further shown that Kronecker’s elimination theory does not correspond to decomposition into primary ideals. (E. N.)

(that is, all determinants of order λ formed from the coefficient matrix of any λ linear forms from $\mathfrak{A}$). If ρ is the rank of $\mathfrak{A}$, so that $d_1, \dots, d_\rho$ are the non-zero determinantal divisors, then the *elementary divisors* of $\mathfrak{A}$ are defined by $e_1 = d_1$, $e_2 = d_2/d_1, \dots, e_\rho = d_\rho/d_{\rho-1}$, $e_{\rho+i} = 0$. It is known that $\mathfrak{A}$ is a finite module having a module basis of exactly ρ linear forms $a_1(\xi), \dots, a_\rho(\xi)$, by which every linear form from $\mathfrak{A}$ can be represented linearly, with integral quantities as coefficients: $\mathfrak{A} = (a_1(\xi), \dots, a_\rho(\xi))$. If A denotes the coefficient matrix of these linear forms, then the determinantal and elementary divisors of $\mathfrak{A}$ agree with those of A ; hence first it follows that *each elementary divisor e_i divides the following one e_{i+1}* . The matrix equation furnished by the elementary-divisor theory,

$$PAQ = \begin{pmatrix} e_1 & & & 0 \\ & e_2 & & \\ & & \ddots & \\ 0 & & & e_\rho \end{pmatrix},$$

further shows – if new indeterminates $\eta_1, \dots, \eta_k$ are introduced by the unimodular transformation $\xi = Q(\eta)$ – that the module equation holds:

$$\mathfrak{A} = (e_1\eta_1, e_2\eta_2, \dots, e_\rho\eta_\rho). \quad (1)$$

The concept most essential for what follows is that of the fundamental module⁵, according to

Definition I. A module $\mathfrak{G}$ is called a *fundamental module* if it has no proper divisor of the same rank⁶.

Thus $\mathfrak{G}$ is a fundamental module if and only if from

$$c \cdot g(\xi) \equiv 0(\mathfrak{G}); \quad c \neq 0 \quad \text{there always follows:} \quad g(\xi) \equiv 0(\mathfrak{G}). \quad (2)$$

The connection between an arbitrary module and a fundamental module is given by

Theorem I. Every module $\mathfrak{A}$ is divisible by one and only one fundamental module $\mathfrak{G}$ of the same rank; $\mathfrak{G}$ is defined as the *smallest divisor of $\mathfrak{A}$ having the same rank, and is represented by*

$$\mathfrak{G} = (\eta_1, \dots, \eta_\rho); \quad (3)$$

$\mathfrak{G}$ shall be called the *fundamental module of $\mathfrak{A}$* .

The condition that $\mathfrak{G}$ is to be the smallest divisor of the same rank is expressed as follows: $\mathfrak{G}$ contains all and only the linear forms $g(\xi)$ that satisfy a relation

$$b \cdot g(\xi) \equiv 0(\mathfrak{A}); \quad b \neq 0. \quad (4)$$

It follows first that $\mathfrak{G}$ is a module, since together with

$$b_1 g_1(\xi) \equiv 0(\mathfrak{A}); \quad b_1 \neq 0; \quad b_2 g_2(\xi) \equiv 0(\mathfrak{A}); \quad b_2 \neq 0$$

also

$$b_1 b_2 (g_1(\xi) - g_2(\xi)) \equiv 0(\mathfrak{A}); \quad b_1 b_2 \neq 0$$

is fulfilled; and of course also $b_1 \cdot c g_1(\xi) \equiv 0(\mathfrak{A})$. But $\mathfrak{G}$ is also a fundamental module; for from

$$c \cdot g(\xi) \equiv 0(\mathfrak{G}); \quad c \neq 0$$

there follows, by (4),

$$bcg(\xi) \equiv 0(\mathfrak{A}); \quad bc \neq 0,$$

and hence again by (4) also $g(\xi) \equiv 0(\mathfrak{G})$.

There is moreover no fundamental module $\mathfrak{G}_1$ different from the smallest divisor $\mathfrak{G}$ of the same rank that satisfies the conditions. For $\mathfrak{G}_1$ would have to be divisible by $\mathfrak{G}$, but as a fundamental module it has no proper divisor of the same rank, and hence is identical with $\mathfrak{G}$. Finally, the module represented by the right hand side of (3) is a fundamental module, since every proper divisor must contain a further indeterminate η , and so has higher rank. Thus (3) gives the uniquely defined fundamental module of $\mathfrak{A}$, and *Theorem I is proved in all its parts.*

⁵Cf. Steinitz, Rechteckige Systeme und Moduln in algebraischen Zahlkörpern II. Math. Ann. 72 (1912), pp. 297–345, No. 36. Hentzelt seems in any case, independently of Steinitz – with whom points of contact are also found elsewhere – to have reached, for his simpler case, the concept in the form of formula (4). (E. N.)

⁶Divisor understood in the module sense: $\mathfrak{A}$ is divisible by $\mathfrak{B}$, $\mathfrak{A} \equiv 0(\mathfrak{B})$, if every element of $\mathfrak{A}$ is contained in $\mathfrak{B}$; $\mathfrak{B}$ is called a proper divisor if it contains elements different from those of $\mathfrak{A}$.

A further concept essential for what follows is Dedekind's concept of the quotient of two modules⁷:

Definition II. The quotient $\mathfrak{c} = \mathfrak{A}/\mathfrak{B}$ of two modules of linear forms is defined as the totality of the integral quantities c satisfying the condition

$$c \cdot \mathfrak{B} \equiv 0(\mathfrak{A}).$$

$\mathfrak{c}$ is an ideal of integral quantities, hence a principal ideal. Correspondingly, the quotient $\mathfrak{C} = \mathfrak{A}/\mathfrak{b}$ of a module $\mathfrak{A}$ of linear forms by an ideal $\mathfrak{b}$ of integral quantities is defined as the totality of the linear forms $c(\xi)$ satisfying the condition

$$c(\xi) \cdot \mathfrak{b} \equiv 0(\mathfrak{A}).$$

$\mathfrak{C}$ is again a module of linear forms, and indeed, as soon as $\mathfrak{b}$ is different from the zero ideal, a module of the same rank as $\mathfrak{A}$.

One need only remark that by definition it is clear that the module property belongs to the quotient in each case; systems of integral quantities with the module property, however, are ideals.

The quotient concept leads to a connection between the fundamental module and the highest elementary divisor, namely:

Theorem II. Between the fundamental module and the highest elementary divisor of $\mathfrak{A}$ there is the reciprocal relation

$$(e_\rho) = \mathfrak{A}/\mathfrak{G}; \quad \mathfrak{G} = \mathfrak{A}/(e_\rho), \quad (5)$$

where (e_ρ) denotes the principal ideal derived from e_ρ .

For since every elementary divisor divides e_ρ , from (1) and (3) we have

$$e_\rho \mathfrak{G} \equiv 0(\mathfrak{A}) \quad \text{and consequently} \quad (e_\rho) \cdot \mathfrak{G} \equiv 0(\mathfrak{A}); \quad (6)$$

hence (e_ρ) is divisible by the quotient $\mathfrak{A}/\mathfrak{G}$. But the converse also holds; for from

$$b\mathfrak{G} \equiv 0(\mathfrak{A}) \quad \text{there follows} \quad b\eta_\rho \equiv 0(\mathfrak{A}) \quad \text{and hence} \quad b \equiv 0((e_\rho)),$$

which proves the first formula (5)⁸. Formula (6) further shows that $\mathfrak{G}$ is divisible by the quotient $\mathfrak{A}/(e_\rho)$; this quotient is a divisor of $\mathfrak{A}$ of the same rank, hence, by the definition of the fundamental module of $\mathfrak{A}$, conversely divisible by $\mathfrak{G}$. Thus the second formula (5) is also proved.

If d denotes any determinant of order ρ from $\mathfrak{A}$ which does not vanish identically, then d is divisible by the ρ -th determinantal divisor d_ρ , and hence by e_ρ ; consequently $d\mathfrak{G} \equiv 0(\mathfrak{A})$, and by the preceding conclusion $\mathfrak{G} = \mathfrak{A}/(d)$. For what follows, however, it is essential that this latter quotient representation for $\mathfrak{G}$ can be proved without going back to the elementary divisors, and hence has a more general validity, according to

Theorem III. If by integral quantities one understands polynomials in several indeterminates⁹, then the quotient representation still holds

$$\mathfrak{G} = \mathfrak{A}/(d), \quad (7)$$

where d denotes any determinant of order ρ from $\mathfrak{A}$ which does not vanish identically.

First it is clear that the notions of rank, fundamental module and module quotient – which now is just no longer a principal ideal – are preserved under this extension, and so is Theorem I, with the exception of the basis representation.

⁷In Dedekind the matter concerns modules of numbers, for which multiplication is also defined, so that Dedekind's quotient does not agree with the quotient defined here. (E. N.)

⁸Set $\mathfrak{A}_0 = \mathfrak{A}, \dots, \mathfrak{A}_\lambda = (\mathfrak{A}, \eta_\rho, \dots, \eta_{\rho-\lambda+1})$, where in each case $\mathfrak{A}_i$ has highest elementary divisor $e_{\rho-i}$. Then by the same conclusions one obtains

$$(e_\rho) = \mathfrak{A}_0/\mathfrak{A}_1, \dots, (e_{\rho-\lambda}) = \mathfrak{A}_\lambda/\mathfrak{A}_{\lambda+1}, \dots, (e_1) = \mathfrak{A}_{\rho-1}/\mathfrak{A}_\rho.$$

More generally, set $\zeta_\lambda = b_{\lambda 1}\eta_1 + \dots + b_{\lambda \lambda}\eta_\lambda$, assume $(b_{\lambda \lambda}, e_\lambda) = 1$, and let

$$\mathfrak{B}_0 = \mathfrak{A}, \dots, \mathfrak{B}_\lambda = (\mathfrak{A}, \zeta_\rho, \dots, \zeta_{\rho-\lambda+1}).$$

Then $\mathfrak{B}_i$ also has highest elementary divisor $e_{\rho-i}$, and hence

$$(e_{\rho-\lambda}) = \mathfrak{B}_\lambda/\mathfrak{B}_{\lambda+1}.$$

⁹In fact only this is used: that the integral quantities reproduce themselves under addition, subtraction and multiplication, with the usual rules of calculation holding, so that they form a ring; and further, that in this ring a product vanishes only when one factor vanishes, so that the ring may be extended to a field by quotient formation (adjunction of pairs of elements). On the other hand no basis representation of the module is used; thus (7) holds for example also for the domain of all algebraic integers as coefficients. (E. N.)

Let now

$$a_1(\xi) = a_{11}\xi_1 + \cdots + a_{1\rho}\xi_\rho, \dots, a_\rho(\xi) = a_{\rho 1}\xi_1 + \cdots + a_{\rho\rho}\xi_\rho$$

be ρ linearly independent linear forms from $\mathfrak{A}$, which in general will not form a module basis. Let the indeterminates ξ be so designated that

$$d = |a_{ij}| \neq 0; \quad i, j = 1, 2, \dots, \rho.$$

It follows that ρ linear forms of special shape belong to $\mathfrak{A}$:

$$d\xi_\lambda + b_{\lambda, \rho+1}\xi_{\rho+1} + \cdots + b_{\lambda, k}\xi_k \equiv 0(\mathfrak{A}) \quad (\lambda = 1, 2, \dots, \rho). \quad (8)$$

But $\mathfrak{A}$ can contain no linear form depending only on $\xi_{\rho+1}, \dots, \xi_k$, say $c_{\rho+1}\xi_{\rho+1} + \cdots + c_k\xi_k$, since otherwise at least one determinant of order $\rho + 1$, namely $d \cdot c_\alpha$, would be non-zero.

Now the quotient $\mathfrak{A}/(d)$, as a divisor of $\mathfrak{A}$ of the same rank, is divisible by $\mathfrak{G}$; but the converse also holds. For every $g(\xi) \equiv 0(\mathfrak{G})$ satisfies, by definition,

$$c \cdot g(\xi) \equiv 0(\mathfrak{A}); \quad c \neq 0 \quad \text{and consequently} \quad d \cdot c \cdot g(\xi) \equiv 0(\mathfrak{A}).$$

By (8), however,

$$d \cdot g(\xi) = g_{\rho+1}\xi_{\rho+1} + \cdots + g_k\xi_k \equiv 0(\mathfrak{A}),$$

and consequently

$$cg_{\rho+1}\xi_{\rho+1} + \cdots + cg_k\xi_k \equiv 0(\mathfrak{A}).$$

By what has just been observed this implies $c \cdot g_{\rho+1} = 0, \dots, c \cdot g_k = 0$, and because $c \neq 0$, also $g_{\rho+1} = 0, \dots, g_k = 0$; hence $d \cdot g(\xi) \equiv 0(\mathfrak{A})$, proving (7).

It should be noted that $d \cdot g(\xi) \equiv 0(\mathfrak{A})$ also follows directly from the fact that the two modules $(a_1(\xi), \dots, a_\rho(\xi))$ and $(g(\xi), a_1(\xi), \dots, a_\rho(\xi))$ have the same rank ρ , so that – putting $g(\xi) = c_1\xi_1 + \cdots + c_\rho\xi_\rho$ – the determinant of order $\rho + 1$

$$\begin{vmatrix} a_1(\xi) & a_{11} & \cdots & a_{1\rho} \\ \vdots & \vdots & & \vdots \\ a_\rho(\xi) & a_{\rho 1} & \cdots & a_{\rho\rho} \\ g(\xi) & c_1 & \cdots & c_\rho \end{vmatrix}$$

vanishes. The proof given above, however, recurs in § 4 following formula (30).

§ 2. Decomposition Theorems for Norms and Modules.

We now return to modules of linear forms with respect to rational integers or polynomials in one indeterminate with coefficients from P .

Definition III. *If, as usual, all linear forms in ξ that are congruent to one another modulo $\mathfrak{A}$ are collected into one residue class, then the symbol $\mathfrak{G} \mid \mathfrak{A}$ shall denote the system of residue classes of $\mathfrak{G}$ modulo $\mathfrak{A}$, that is, the system of all residue classes modulo $\mathfrak{A}$ that consist of elements of $\mathfrak{G}$. The norm of $\mathfrak{G}$ with respect to $\mathfrak{A}$ – in symbols $N(\mathfrak{G} \mid \mathfrak{A})$ – is defined as the determinant of the transition substitution from $\mathfrak{G}$ to $\mathfrak{A}$, that is, as the determinant of the substitution which expresses any linearly independent module basis of $\mathfrak{A}$ by such a basis of the fundamental module $\mathfrak{G}$ of $\mathfrak{A}$.*

From (1) and (3), respectively from the remark to Theorem II, one obtains:

$$N(\mathfrak{G} \mid \mathfrak{A}) = e_1 e_2 \cdots e_\rho = d_\rho; \quad \mathfrak{G} = \mathfrak{A}/(N(\mathfrak{G} \mid \mathfrak{A})). \quad (9)$$

From (1), (3) and (9) it follows in the familiar way that, in the case of rational integers, $N(\mathfrak{G} \mid \mathfrak{A})$ represents the number of residue classes of $\mathfrak{G}$ modulo $\mathfrak{A}$, whereas in the case of polynomials in one indeterminate – where the number of residue classes is in general infinite – the degree of $N(\mathfrak{G} \mid \mathfrak{A})$ becomes equal to the *number of residue classes linearly independent with respect to P* . If $\mathfrak{B}$ is a divisor of $\mathfrak{A}$ of the same rank, then, together with a representative of any such residue class, $\mathfrak{B}$ also contains all elements of the residue class; hence $\mathfrak{B}$ arises from $\mathfrak{A}$ by adjoining finitely many residue classes, or their linear combinations. It follows at once that:

Theorem IV. *If $\mathfrak{B}$ is a divisor of $\mathfrak{A}$ of the same rank, so that the fundamental modules agree, and if moreover $N(\mathfrak{G} \mid \mathfrak{B}) = N(\mathfrak{G} \mid \mathfrak{A})$, then also $\mathfrak{B} = \mathfrak{A}$.*

On the connection between the decomposition of norms and modules one has

Theorem V. *Let*

$$N(\mathfrak{G} \mid \mathfrak{A}) = r \cdot s, \quad (r, s) = 1; \quad (10)$$

then there exists one and only one module $\mathfrak{R}$, and likewise one and only one module $\mathfrak{S}$, both divisors of $\mathfrak{A}$ of the same rank, such that

$$r = N(\mathfrak{G} \mid \mathfrak{R}), \quad s = N(\mathfrak{G} \mid \mathfrak{S})$$

holds. $\mathfrak{R}$ and $\mathfrak{S}$ are defined by

$$\mathfrak{R} = \mathfrak{A}/(s); \quad \mathfrak{S} = \mathfrak{A}/(r),$$

and $\mathfrak{A}$ is equal to the least common multiple of $\mathfrak{R}$ and $\mathfrak{S}$.¹⁰ If one sets $r_\rho = (r, e_\rho)$, $s_\rho = (s, e_\rho)$, then the reciprocities hold

$$(s_\rho) = \mathfrak{A}/\mathfrak{R}; \quad \mathfrak{R} = \mathfrak{A}/(s_\rho); \quad (r_\rho) = \mathfrak{A}/\mathfrak{S}; \quad \mathfrak{S} = \mathfrak{A}/(r_\rho). \quad (11)$$

Indeed, setting generally $r_i = (r, e_i)$, $s_i = (s, e_i)$, one obtains from (9) and (10):

$$(r_i, s_i) = 1; \quad e_i = r_i s_i; \quad r = r_1 \cdots r_\rho; \quad s = s_1 \cdots s_\rho,$$

and each r_i respectively s_i divides the following r_{i+1} respectively s_{i+1} . Thus, if one sets

$$\mathfrak{R} = (r_1 \eta_1, \dots, r_\rho \eta_\rho); \quad \mathfrak{S} = (s_1 \eta_1, \dots, s_\rho \eta_\rho),$$

then $\mathfrak{R}$ and $\mathfrak{S}$ are divisors of $\mathfrak{A}$ of the same rank; owing to the relative primeness of r_i and s_i , $\mathfrak{A}$ becomes the least common multiple of $\mathfrak{R}$ and $\mathfrak{S}$, and moreover

$$N(\mathfrak{G} \mid \mathfrak{R}) = r_1 \cdots r_\rho = r; \quad N(\mathfrak{G} \mid \mathfrak{S}) = s_1 \cdots s_\rho = s.$$

Furthermore

$$s_\rho \mathfrak{R} \equiv 0(\mathfrak{A}); \quad r_\rho \mathfrak{S} \equiv 0(\mathfrak{A});$$

and from $c\mathfrak{R} \equiv 0(\mathfrak{A})$ follows $cr_\rho \eta_\rho \equiv 0(\mathfrak{A})$, hence $c \equiv 0((s_\rho))$, proving $(s_\rho) = \mathfrak{A}/\mathfrak{R}$, and similarly $(r_\rho) = \mathfrak{A}/\mathfrak{S}$. From

$$b(\eta) = b_1 \eta_1 + \cdots + b_\rho \eta_\rho \equiv 0(\mathfrak{A}/(s_\rho))$$

it further follows, because all r_i are relatively prime to s_ρ , that $b_i \equiv 0((r_i))$; hence $\mathfrak{R} = \mathfrak{A}/(s_\rho)$ and similarly $\mathfrak{S} = \mathfrak{A}/(r_\rho)$. Thus the reciprocities (11) are proved; in the special case $r = 1$ they pass into (5).

It remains to show the *unique determination* of $\mathfrak{R}$ and $\mathfrak{S}$. Let $\overline{\mathfrak{R}}$ and $\overline{\mathfrak{S}}$ be modules satisfying the conditions of Theorem V, and let $\bar{r}_i$ and $\bar{s}_i$ be their elementary divisors. Since $\bar{r}_i$ and $\bar{s}_i$ are divisors of e_i , it follows, because

$$r = \bar{r}_1 \cdots \bar{r}_\rho, \quad s = \bar{s}_1 \cdots \bar{s}_\rho, \quad (\bar{r}_i, \bar{s}_i) = 1,$$

that also

$$e_i = \bar{r}_i \bar{s}_i, \quad \bar{r}_i = (r, e_i) = r_i, \quad \bar{s}_i = (s, e_i) = s_i.$$

Thus $\overline{\mathfrak{R}}$ and $\overline{\mathfrak{S}}$ agree with $\mathfrak{R}$ and $\mathfrak{S}$ in elementary divisors and in fundamental module. If therefore, by a unimodular substitution $\eta = Q(\xi)$, new indeterminates ξ are introduced, then

$$\overline{\mathfrak{R}} = (r_1 \xi_1, \dots, r_\rho \xi_\rho);$$

$$\mathfrak{A} = (r_1 s_1 (q_{11} \xi_1 + \cdots + q_{1\rho} \xi_\rho), \dots, r_\rho s_\rho (q_{\rho 1} \xi_1 + \cdots + q_{\rho \rho} \xi_\rho)).$$

From $\mathfrak{A} \equiv 0(\mathfrak{R})$ one thus obtains

$$r_i s_i (q_{i1} \xi_1 + \cdots + q_{i\rho} \xi_\rho) \equiv 0(\mathfrak{R});$$

therefore $r_i s_i q_{ij} \equiv 0((r_i))$. Since $(r, s) = 1$, this gives $r_i q_{ij} \equiv 0((r_i))$, or $\overline{\mathfrak{R}} \equiv 0(\mathfrak{R})$. But since $N(\mathfrak{G} \mid \mathfrak{R}) = N(\mathfrak{G} \mid \overline{\mathfrak{R}})$, Theorem IV gives $\mathfrak{R} = \overline{\mathfrak{R}}$, and correspondingly $\mathfrak{S} = \overline{\mathfrak{S}}$. Theorem V is thereby proved in all its parts.

¹⁰Least common multiple $[\mathfrak{R}, \mathfrak{S}]$ understood in the module sense: the totality of linear forms that are divisible both by $\mathfrak{R}$ and by $\mathfrak{S}$. Correspondingly, the greatest common divisor $(\mathfrak{R}, \mathfrak{S})$ in the module sense is to be understood as the totality of linear forms that can be represented as the sum of an element from $\mathfrak{R}$ and an element from $\mathfrak{S}$.

§ 3. Ideals of Polynomials.

Let $\bar{\mathfrak{m}}$ be an *ideal of polynomials* in the n indeterminates $y_1, \dots, y_n$, with coefficients from an arbitrary, abstractly defined field P ; that is, together with $\Phi_1(y)$ and $\Phi_2(y)$, $\bar{\mathfrak{m}}$ also contains the difference $\Phi_1(y) - \Phi_2(y)$, and together with $\Phi(y)$ it also contains $A(y) \cdot \Phi(y)$, where $A(y)$ denotes an arbitrary polynomial with coefficients from P ¹¹.

Let the y be subjected to a transformation with indeterminates as coefficients,

$$y = U(x); \quad \text{for instance} \quad \begin{cases} y_1 = x_1, \\ y_2 = u_{21}x_1 + x_2, \\ \dots \\ y_n = u_{n1}x_1 + \dots + u_{n,n-1}x_{n-1} + x_n, \end{cases} \quad (12)$$

and adjoin the indeterminates $u_{\mu\nu}$ to the coefficient domain of the polynomials, which thereby passes into a field $P(u)$. By this adjunction let $\bar{\mathfrak{m}}$ pass into $\bar{\mathfrak{m}}_{(u)}$; hence $\bar{\mathfrak{m}}_{(u)}$ contains, besides the polynomials $\Phi_\lambda(y)$ from $\bar{\mathfrak{m}}$, all linear combinations $\sum \alpha_\lambda(u)\Phi_\lambda(y)$ of finitely many Φ_λ , where $\alpha_\lambda(u)$ denotes rational functions of the $u_{\mu\nu}$ with coefficients from P . By multiplication with a suitable polynomial in u , the $\alpha_\lambda(u)$ may, without restriction of generality, be assumed to be power products in u . Thus one has

$$\sum \alpha_\lambda(u)\Phi_\lambda(y) \equiv 0(\bar{\mathfrak{m}}_{(u)}); \quad \Phi_\lambda(y) \equiv 0(\bar{\mathfrak{m}}) \quad \text{and conversely.} \quad (13)$$

By means of (12), $\bar{\mathfrak{m}}_{(u)}$ passes into an ideal $\mathfrak{m}$ of polynomials in $x_1, \dots, x_n$ with coefficients from $P(u)$:

$$\bar{\mathfrak{m}}_{(u)} = \mathfrak{m} \quad \text{by means of} \quad y = U(x). \quad (14)$$

The combination of the relations (14) and (13) says the following:

$$\text{From } F(x) \equiv 0(\mathfrak{m}); \quad F(x) = \Phi(y) = \sum \alpha_\lambda(u)\Phi_\lambda(y) = \sum \alpha_\lambda(u)F_\lambda(x) \text{ follows } F_\lambda(x) \equiv 0(\mathfrak{m}); \quad (15)$$

whereas (14) and (15) conversely yield (13) again.

Definition IV. Ideals $\mathfrak{m}$ of polynomials in $x_1, \dots, x_n$ with coefficients from $P(u)$ which satisfy the relations (15), and hence arise from $\bar{\mathfrak{m}}_{(u)}$ by means of $y = U(x)$, shall be called *transformed ideals*.

Transformed ideals are distinguished by the existence of regular polynomials; a polynomial of degree r is called *regular with respect to x_i* if it contains the term x_i^r with non-zero coefficient. One sees that the polynomials $F_i(x)$ are regular with respect to x_1 . In what follows the main point will be to regard these transformed ideals as modules of linear forms in certain power products of the x . For this the concept of the fundamental ideal must be fixed; later it will pass into the fundamental module. We first give a notation that will be maintained throughout from now on:

Notation. By $F^{(i)}, f^{(i)}, a^{(i)}, \dots$ we shall always mean polynomials in the x that are free of $x_1, \dots, x_{i-1}$.

Definition V. To every ideal $\mathfrak{m}$ there are defined n fundamental ideals $\mathfrak{g}_0, \mathfrak{g}_1, \dots, \mathfrak{g}_{n-1}$ by the following stipulation: the fundamental ideal of stage $(i-1)$, $\mathfrak{g}_{i-1}$, contains all and only the polynomials $G(x)$ for which there is a polynomial $b^{(i)}$ – in general varying with $G(x)$ – such that

$$b^{(i)}G(x) \equiv 0(\mathfrak{m}); \quad b^{(i)} \neq 0. \quad (16)$$

That the $\mathfrak{g}$ are in fact ideals corresponds exactly to the proof of the module property for $\mathfrak{G}$ following formula (4), with (16) playing the analogous role. The ideal $\mathfrak{g}_i$ is divisible by $\mathfrak{g}_{i-1}$, since every polynomial $b^{(i+1)}$ is at the same time a $b^{(i)}$.

For the fundamental ideals one has:

Theorem VI. The fundamental ideals of transformed ideals are transformed ideals¹².

The proof rests on a known theorem of Dedekind–Mertens¹³: let a and b be indeterminates, and set

$$\sum a_{i_1, \dots, i_s} z_1^{i_1} \dots z_s^{i_s} \cdot \sum b_{j_1, \dots, j_s} z_1^{j_1} \dots z_s^{j_s} = \sum c_{k_1, \dots, k_s} z_1^{k_1} \dots z_s^{k_s},$$

¹¹The conceptually resulting designation “ideal” instead of the formerly generally customary “module” or “module of forms” already proves necessary here in order to distinguish these from the modules of linear forms. (E. N.)

¹²Theorem VI is used only in § 7.

¹³Dedekind: Über einen arithmetischen Satz von Gauß, Mitt. d. deutsch. math. Ges. zu Prag 1892. F. Mertens: Über einen algebraischen Satz, Ber. d. Ak. d. Wissensch. Wien 101 (1892), pp. 1560–1566. – Kronecker’s extension of Gauss’s theorem to polynomials with indeterminate coefficients is an immediate consequence of this theorem.

$$\sum i \leq \nu, \quad \sum j \leq \mu, \quad \sum k \leq \nu + \mu.$$

If $\mathfrak{A}, \mathfrak{B}, \mathfrak{C}$ denote the modules derived respectively from the a, b, c by means of rational integers, then there exists an exponent q such that the identity holds:

$$\mathfrak{A}^q \mathfrak{B} = \mathfrak{A}^{q-1} \mathfrak{C}. \quad (17)$$

Here $\mathfrak{A}^q$ consists of the power products of q -th dimension in the a and their integral linear combinations.

For the proof of Theorem VI, now put

$$\begin{cases} G(x) = \Gamma(y) = \sum \alpha_\lambda(u) \Gamma_\lambda(y) = \sum \alpha_\lambda(u) G_\lambda(x), \\ b^{(i)}(x) = \varphi(y) = \sum \beta_\mu(u) \varphi_\mu(y), \end{cases} \quad (18)$$

where $\alpha_\lambda(u)$ and $\beta_\mu(u)$ are power products in the u . Then, by (16), (14) and (18),

$$\sum \beta_\mu(u) \varphi_\mu(y) \cdot \sum \alpha_\lambda(u) \Gamma_\lambda(y) \equiv 0(\overline{\mathfrak{m}}_{(u)}); \quad \sum \beta_\mu(u) \varphi_\mu(y) \neq 0. \quad (19)$$

On the left stands the product of two polynomials in u whose coefficients are the polynomials $\varphi_\mu(y)$ and $\Gamma_\lambda(y)$; the coefficients of the product polynomial in u are, by the right hand side of (19), polynomials from $\overline{\mathfrak{m}}$. If, therefore, in (17) one replaces the a, b, c by these polynomials in y , one obtains

$$\left\{ \sum \beta_\mu(u) \varphi_\mu(y) \right\}^q \cdot \Gamma_\lambda(y) \equiv 0(\overline{\mathfrak{m}}_{(u)}),$$

which by (18) is equivalent to:

$$b^{(i)}(x)^q \cdot G_\lambda(x) \equiv 0(\mathfrak{m}); \quad G_\lambda(x) \equiv 0(\mathfrak{g}_{i-1}). \quad (20)$$

The relations (16), (18) and (20) show that the *fundamental ideals* $\mathfrak{g}_{i-1}$ are indeed transformed ideals.